

Cognitive APIs

APIs and Cognitive APIs

- **Cognitive APIs:** APIs that provide a cognitive (data science) service.
- **Multiple cloud providers provide such cognitive APIs:** Microsoft, Amazon, Google, IBM, Twitter, etc.
- **Features of APIs**
 - Input, Output
 - Rate limits
 - Authentication
 - HTTP auth: provide username and password.
 - API keys: a unique generated value is assigned to each first time user
 - OAuth
 - Pricing

Azure Cognitive Service APIs



Phi-3 open models

Build with the most capable and cost-effective small language models (SLMs) available.

[Learn more](#)



Azure AI Content Safety

Monitor text and images to detect offensive or inappropriate content.

[Learn more](#)



Azure AI Vision

Read text, analyze images, and detect faces with optical character recognition (OCR) and machine learning.

[Learn more](#)



Azure OpenAI Service

Build your own copilot and generative AI applications with cutting-edge language and vision models.

[Learn more](#)



Azure AI Translator

Translate documents and text in real time across more than 100 languages.

[Learn more](#)



Azure AI Language

Build conversational interfaces, summarize documents, and analyze text using prebuilt AI-powered features.

[Learn more](#)



Azure AI Search

Retrieve the most relevant data using keyword, vector, and hybrid search.

[Learn more](#)



Azure AI Speech

Use industry-leading AI services such as speech-to-text, text-to-speech, speech translation, and speaker recognition.

[Learn more](#)



Azure AI Document Intelligence

Apply advanced machine learning to extract text, key-value pairs, tables, and structures from documents.

[Learn more](#)

Azure AI Vision



| Image analysis

Image analysis that pulls from more than 10,000 concepts and objects to detect, classify, caption, and generate insights.



| Optical character recognition (OCR)

Optical character recognition (OCR) to extract printed and handwritten text from images with varied languages and writing styles.



| Facial recognition

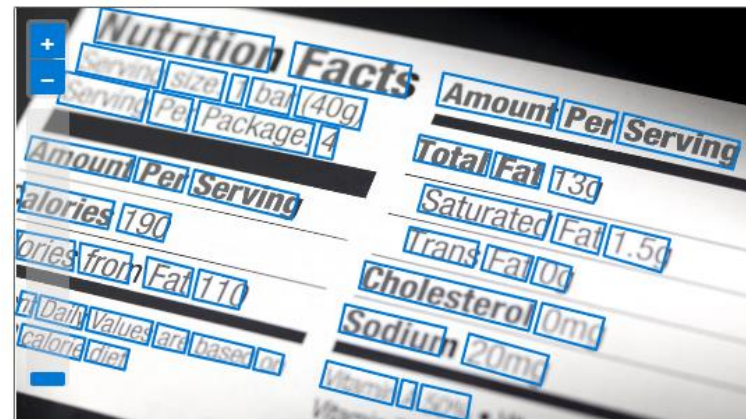
Facial recognition to create intelligent applications that recognize and verify human identity.

Optical Character Recognition

Use one of your own files or choose from a sample below.



Sample form #3



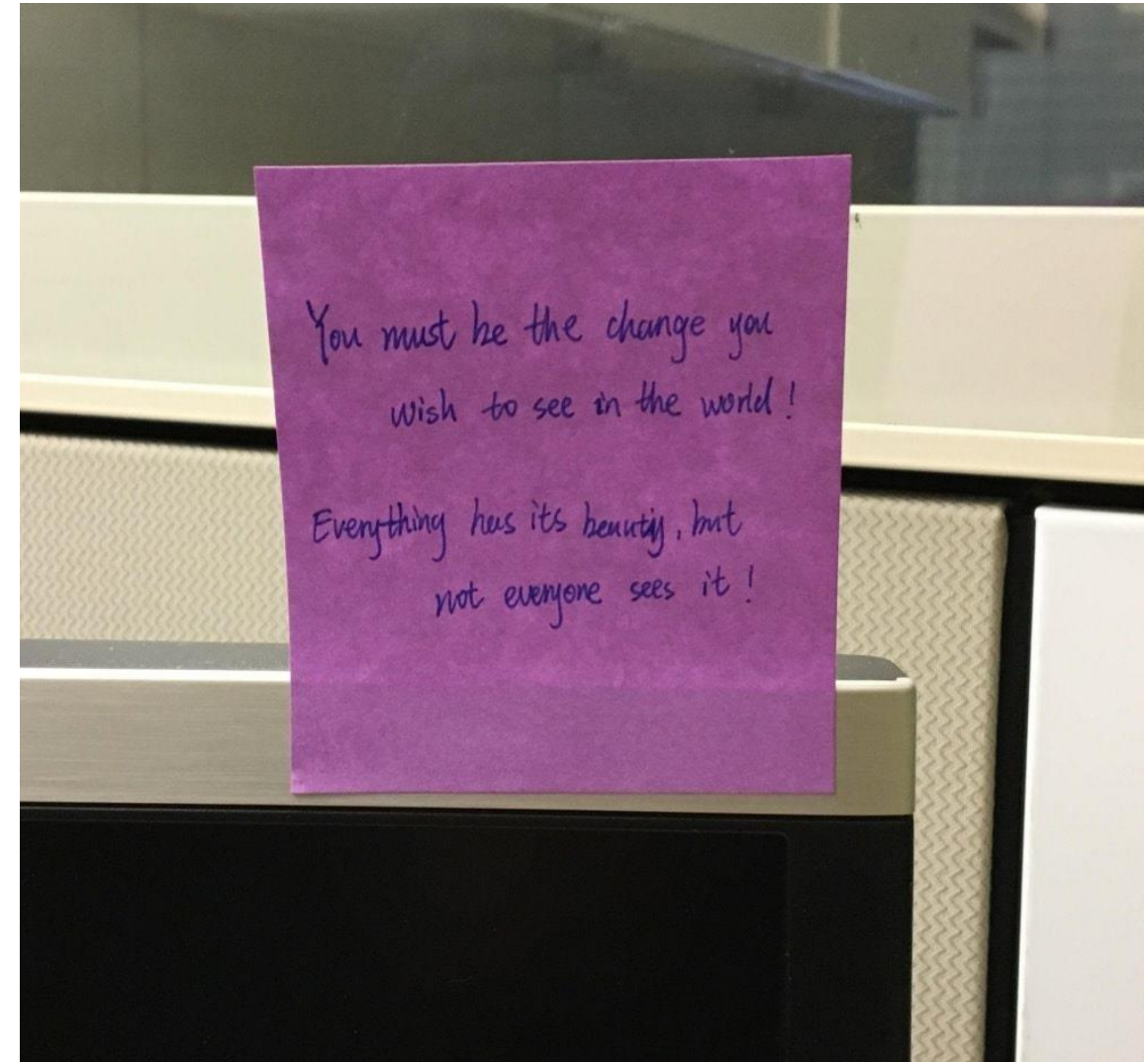
Detected attributes

JSON

```

Nutrition Facts Amount Per Serving
Serving size: 1 bar (40g)
Serving Per Package: 4
Total Fat 13g
Saturated Fat 1.5g
Amount Per Serving
Trans Fat 0g
Calories 190
Calories from Fat 110
% Daily Values are based on
calorie diet.
Cholesterol 0mg
Sodium 20mg
Vitamin A 50%

```



Read handwritten text from images

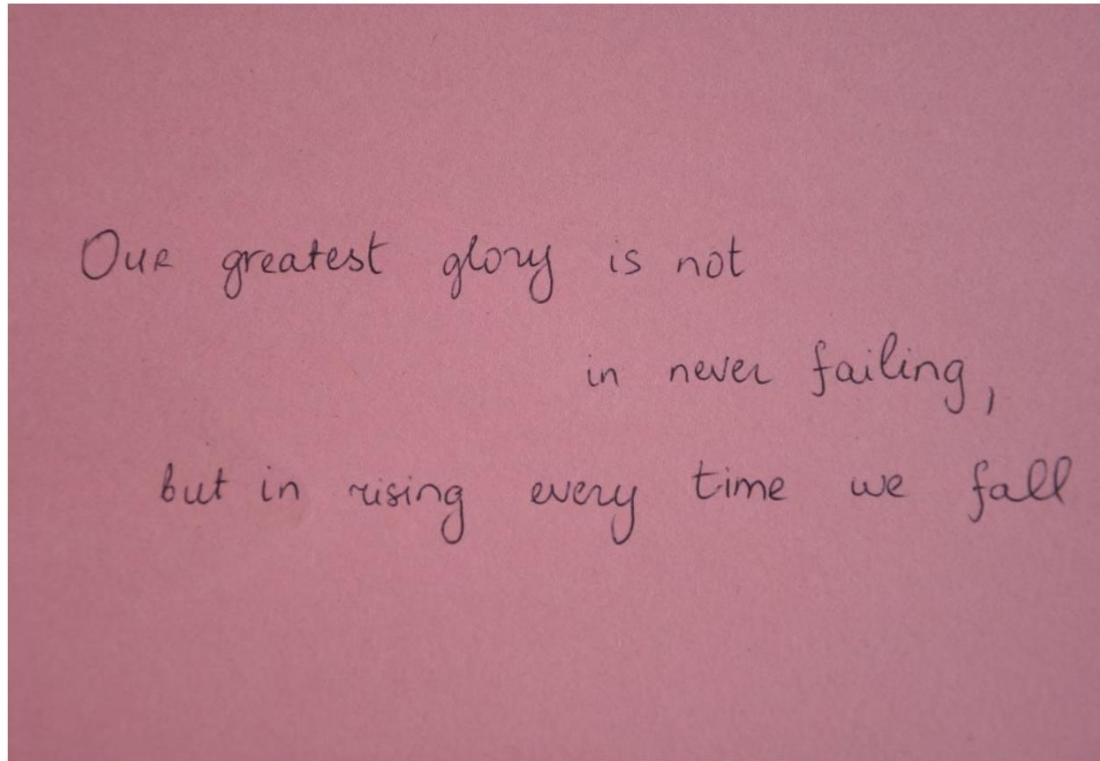


Image URL

Submit

 Browse

Preview

JSON

OUR greatest glory is not
 in never failing,
 om
 but in rising
 _every
 time
 fall

People Detection



```
"peopleResult": {
  "values": [
    {
      "boundingBox": {
        "x": 0,
        "y": 41,
        "w": 95,
        "h": 189
      },
      "confidence": 0.9474349617958069
    },
    {
      "boundingBox": {
        "x": 204,
        "y": 96,
        "w": 95,
        "h": 134
      },
      "confidence": 0.9470965266227722
    }
  ]
}
```

```
{
  "boundingBox": {
    "x": 53,
    "y": 20,
    "w": 136,
    "h": 210
  },
  "confidence": 0.8943784832954407
},
{
  "boundingBox": {
    "x": 170,
    "y": 31,
    "w": 91,
    "h": 199
  },
  "confidence": 0.2713555097579956
}
```

Image Description



```
{
  "description": {
    "tags": [
      "outdoor",
      "city",
      "white"
    ],
    "captions": [
      {
        "text": "a city with tall buildings",
        "confidence": 0.48468858003616333
      }
    ]
  },
  "requestId": "7e5e5cac-ef16-43ca-a0c4-02bd49d379e9",
  "metadata": {
    "height": 300,
    "width": 239,
    "format": "Png"
  },
  "modelVersion": "2021-05-01"
}
```

Object Detection



```
{
  "name": "kitchen appliance",
  "confidence": 0.501,
  "boundingBox": {"x":730,"y":66,"w":135,"h":85}
},
{
  "name": "computer keyboard",
  "confidence": 0.51,
  "boundingBox": {"x":523,"y":377,"w":185,"h":46}
},
{
  "name": "Laptop",
  "confidence": 0.85,
  "boundingBox": {"x":471,"y":218,"w":289,"h":226}
},
{
  "name": "person",
  "confidence": 0.855,
  "boundingBox": {"x":654,"y":0,"w":584,"h":473}
}
```

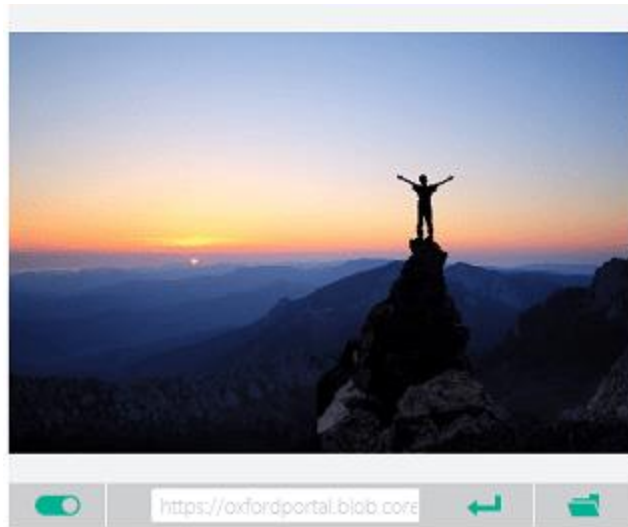
Image tagging



```
{"name": "grass","confidence": 0.996}, {"name": "outdoor","confidence": 0.996}, {"name": "building","confidence": 0.989}, {"name": "property","confidence": 0.985}, {"name": "plant","confidence": 0.979}, {"name": "sky","confidence": 0.976}, {"name": "home","confidence": 0.973}, {"name": "house","confidence": 0.973}, {"name": "real estate","confidence": 0.972}, {"name": "yard","confidence": 0.948}, {"name": "siding","confidence": 0.945}, {"name": "porch","confidence": 0.941}, {"name": "cottage","confidence": 0.914}, {"name": "tree","confidence": 0.911}, {"name": "farmhouse","confidence": 0.899}, {"name": "window","confidence": 0.895}, {"name": "lawn","confidence": 0.894}, {"name": "backyard","confidence": 0.893}, {"name": "garden buildings","confidence": 0.886}, {"name": "roof","confidence": 0.870}, {"name": "driveway","confidence": 0.867}, {"name": "land lot","confidence": 0.856}, {"name": "landscaping","confidence": 0.854}
```


Smart-cropped thumbnails

1. Remove distracting elements from the image and identify the *area of interest*—the area of the image in which the main object(s) appears.
2. Crop the image based on the identified *area of interest*.
3. Change the aspect ratio to fit the target thumbnail dimensions.



Brand detection

Celebrity and Landmark detection

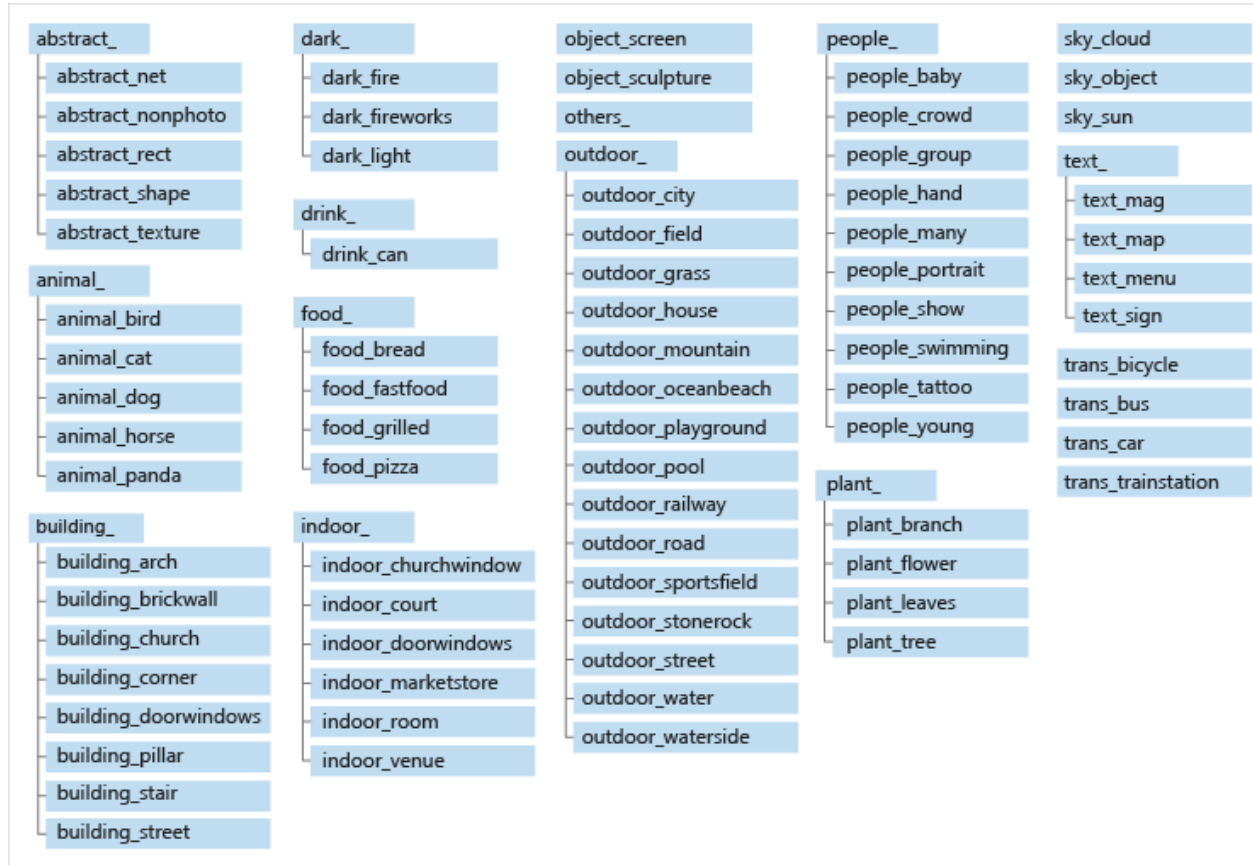


```
"brands": [
  {
    "name": "Microsoft",
    "rectangle": {
      "x": 20,
      "y": 97,
      "w": 62,
      "h": 52
    }
  }
]
```

```
"celebrities": [{
  "faceRectangle": {
    "top": 391,
    "left": 318,
    "width": 184,
    "height": 184
  },
  "name": "Satya Nadella",
  "confidence": 0.99999856948852539
}]
```

Image categorization

86-category taxonomy



people_group



animal_dog

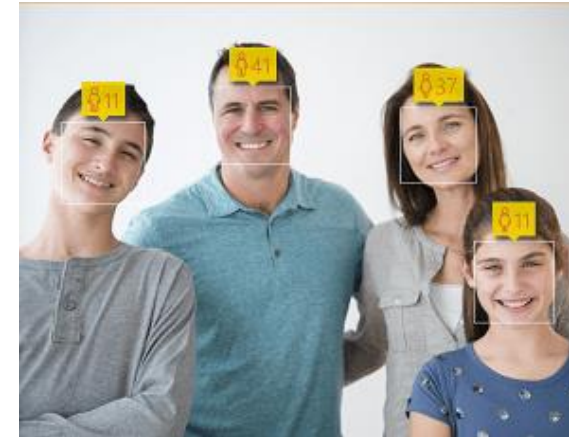


outdoor_mountain



food_bread

Face Detection

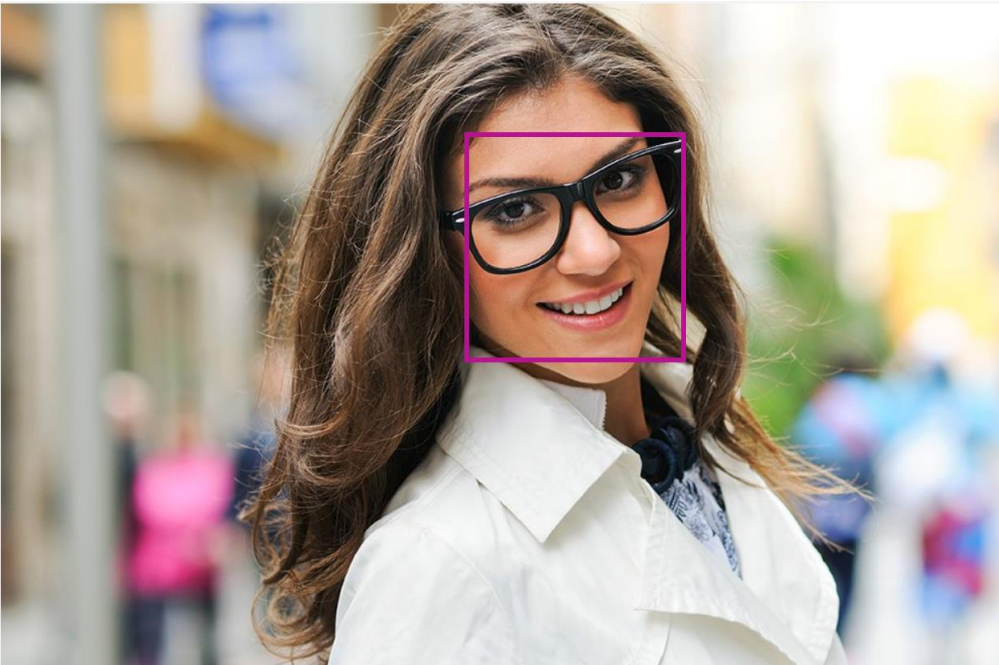


```
"faces": [
  {
    "age": 23,
    "gender": "Female",
    "faceRectangle": {
      "top": 45,
      "left": 194,
      "width": 44,
      "height": 44
    }
  }
],
```

```
"faces": [
  {
    "age": 11,
    "gender": "Male",
    "faceRectangle": {
      "top": 62,
      "left": 22,
      "width": 45,
      "height": 45
    }
  },
  {
    "age": 11,
    "gender": "Female",
    "faceRectangle": {
      "top": 127,
      "left": 240,
      "width": 42,
      "height": 42
    }
  },
],
```

```
{
  "age": 37,
  "gender": "Female",
  "faceRectangle": {
    "top": 55,
    "left": 200,
    "width": 41,
    "height": 41
  }
},
{
  "age": 41,
  "gender": "Male",
  "faceRectangle": {
    "top": 45,
    "left": 103,
    "width": 39,
    "height": 39
  }
}
```


Face Detection



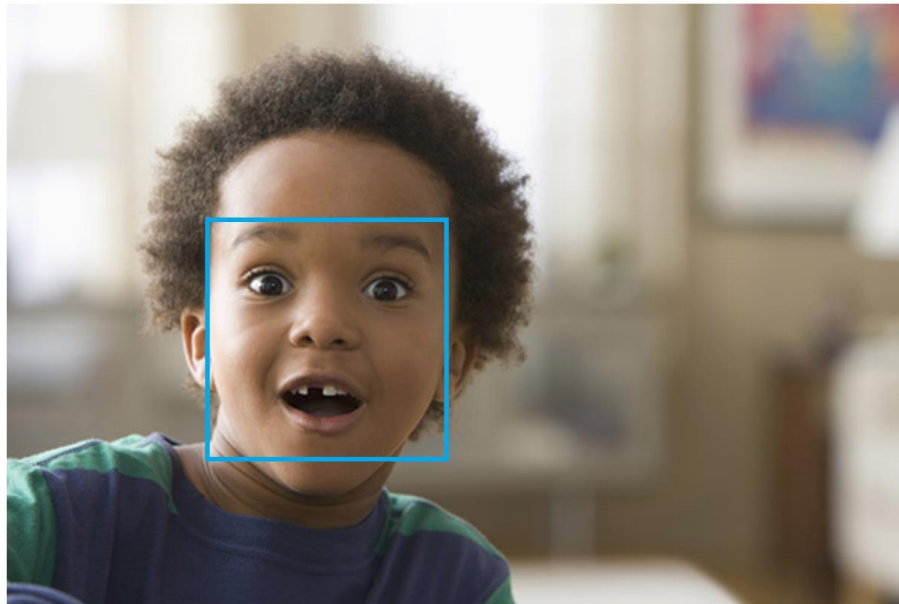
```
JSON: [ {
  "faceId": "add3ab06-6d7f-48ee-94c4-6d97abcfa34d",
  "faceRectangle": { "top": 128, "left": 459, "width": 224, "height": 224 },
  "faceAttributes": {
    "hair": {
      "bald": 0.1,
      "invisible": false,
      "hairColor": [ { "color": "brown", "confidence": 0.99 }, { "color": "black", "confidence": 0.57 }, { "color": "red", "confidence": 0.36 }, {
        "color": "blond", "confidence": 0.34 }, { "color": "gray", "confidence": 0.15 }, { "color": "other", "confidence": 0.13 } ]
    },
    "smile": 1.0,
    "headPose": { "pitch": 0.0, "roll": -16.9, "yaw": 16.7 },
    "gender": "female",
    "age": 24.0,
    "facialHair": { "moustache": 0.0, "beard": 0.0, "sideburns": 0.0 },
    "glasses": "ReadingGlasses",
    "makeup": { "eyeMakeup": true, "lipMakeup": true },
    "emotion": { "anger": 0.0, "contempt": 0.0, "disgust": 0.0, "fear": 0.0, "happiness": 1.0, "neutral": 0.0, "sadness": 0.0, "surprise": 0.0
    },
    "occlusion": { "foreheadOccluded": false, "eyeOccluded": false, "mouthOccluded": false },
    "accessories": [ { "type": "glasses", "confidence": 1.0 } ],
    "blur": { "blurLevel": "low", "value": 0.0 },
    "exposure": { "exposureLevel": "goodExposure", "value": 0.48 },
    "noise": { "noiseLevel": "low", "value": 0.0 }
  },
  "faceLandmarks": {
    "pupilLeft": { "x": 504.8, "y": 206.8 }, "pupilRight": { "x": 602.5, "y": 178.4 }, "noseTip": { "x": 593.5, "y": 247.3 }, "mouthLeft": { "x":
    529.8, "y": 300.5 }, "mouthRight": { "x": 626.0, "y": 277.3 }, "eyebrowLeftOuter": { "x": 461.0, "y": 186.8 }, "eyebrowLeftInner": { "x":
    541.9, "y": 178.9 }, "eyeLeftOuter": { "x": 490.9, "y": 209.0 }, "eyeLeftTop": { "x": 509.1, "y": 199.5 }, "eyeLeftBottom": { "x": 509.3,
    "y": 213.9 }, "eyeLeftInner": { "x": 529.0, "y": 205.0 }, "eyebrowRightInner": { "x": 579.2, "y": 169.2 }, "eyebrowRightOuter": { "x":
    633.0, "y": 136.4 }, "eyeRightInner": { "x": 590.5, "y": 184.5 }, "eyeRightTop": { "x": 604.2, "y": 171.5 }, "eyeRightBottom": { "x":
    608.4, "y": 184.0 }, "eyeRightOuter": { "x": 623.8, "y": 173.7 }, "noseRootLeft": { "x": 549.8, "y": 200.3 }, "noseRootRight": { "x":
    580.7, "y": 192.3 }, "noseLeftAlarTop": { "x": 557.2, "y": 234.6 }, "noseRightAlarTop": { "x": 603.2, "y": 225.1 }, "noseLeftAlarOutTip":
    { "x": 545.4, "y": 255.5 }, "noseRightAlarOutTip": { "x": 615.9, "y": 239.5 }, "upperLipTop": { "x": 591.1, "y": 278.4 },
    "upperLipBottom": { "x": 593.2, "y": 288.7 }, "underLipTop": { "x": 597.1, "y": 308.0 }, "underLipBottom": { "x": 600.3, "y": 324.8 } } ] }
```


Emotion Recognition

Emotion recognition

The Face API now integrates emotion recognition, returning the confidence across a set of emotions for each face in the image such as anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. These emotions are understood to be cross-culturally and universally communicated with particular facial expressions.

See it in action



```

Detection result:
1 faces detected

JSON:
[
  {
    "faceRectangle": {
      "top": 141,
      "left": 130,
      "width": 162,
      "height": 162
    },
    "scores": {
      "anger": 9.29041E-06,
      "contempt": 0.000118981574,
      "disgust": 3.15619363E-05,
      "fear": 0.000589638,
      "happiness": 0.06630674,
      "neutral": 0.00555004273,
      "sadness": 7.44669524E-06,
      "surprise": 0.9273863
    }
  }
]

```

Image URL

Submit

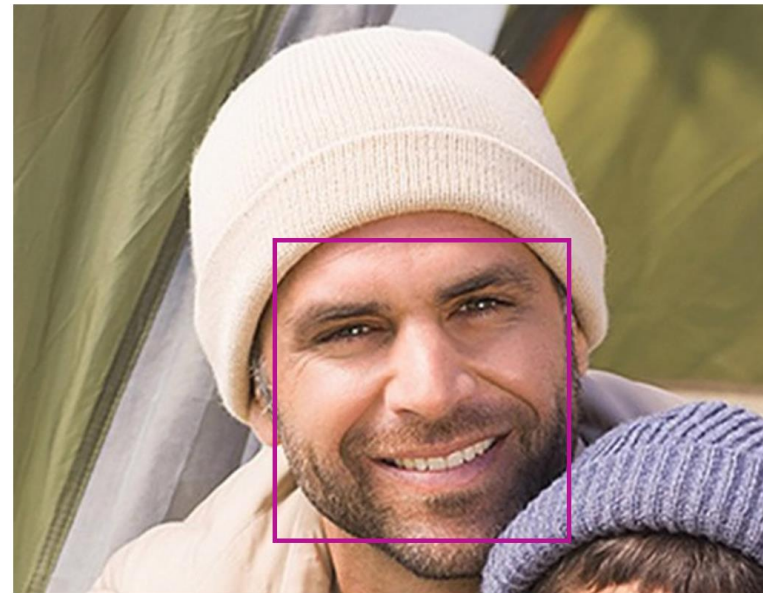
 Browse

Face Verification

Face verification

Check the likelihood that two faces belong to the same person. The API will return a confidence score about how likely it is that the two faces belong to one person.

See it in action



Verification result: The two faces belong to the same person. **Confidence is 0.7349.**

Image Type Detection

- Clip art type and line drawing detection

Value	Meaning
0	Non-clip-art
1	Ambiguous
2	Normal-clip-art
3	Good-clip-art



"clipArtType": 3,
"lineDrawingType": 0



"clipArtType": 0,
"lineDrawingType": 0



"clipArtType": 2,
"lineDrawingType": 1



"clipArtType": 0,
"lineDrawingType": 0

Color scheme detection



Foreground: Black
Background: White
Colors: Black, White, Green

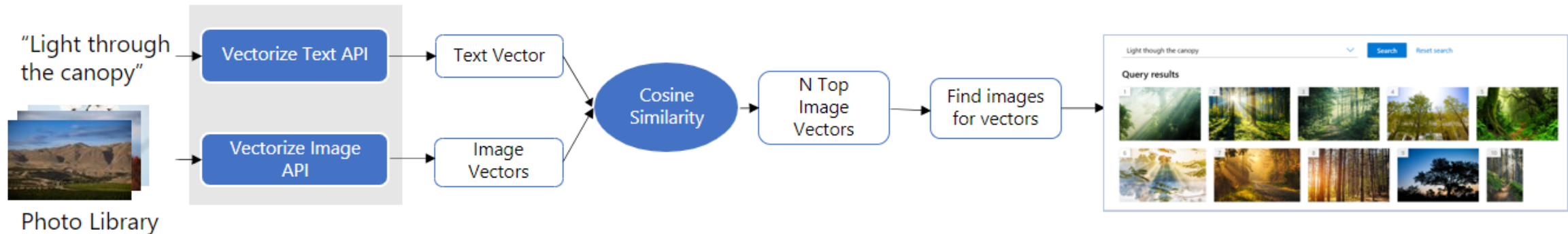
```
{
  "color": {
    "dominantColorForeground": "Black",
    "dominantColorBackground": "Black",
    "dominantColors": ["Black", "White"],
    "accentColor": "BB6D10",
    "isBwImg": false
  },
  "requestId": "0dc394bf-db50-4871-bdcc-13707d9405ea",
  "metadata": {
    "height": 202,
    "width": 300,
    "format": "Jpeg"
  }
}
```



Foreground: Black
Background: Black
Colors: Black

Multimodal embeddings

- Generating a vector representation of an image that captures its features and characteristics.
- Useful for image retrieval.
- Multimodal embedding is not designed to analyze medical images for diagnostic features or disease patterns.



Background removal

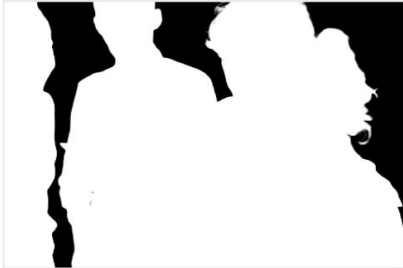
Original image	With background removed	Alpha matte
		
		
		

Image Analysis: Custom models for classification and object detection

- Select images that vary by: camera angle, lighting, background, visual style, individual/grouped subject(s), size, type.
- JPEG, PNG, GIF, BMP, WEBP, ICO, TIFF, or MPO format.
- COCO file

```
"images": [  
  {  
    "id": 1,  
    "width": 500,  
    "height": 828,  
    "file_name": "0.jpg",  
    "absolute_url": "https://blobstorage1.blob.core.windows.net/cpgcontainer/0.jpg"  
  },  
  {  
    "id": 2,  
    "width": 754,  
    "height": 832,  
    "file_name": "1.jpg",  
    "absolute_url": "https://blobstorage1.blob.core.windows.net/cpgcontainer/1.jpg"  
  },  
]
```

```
"annotations": [  
  {  
    "id": 1,  
    "category_id": 7,  
    "image_id": 1,  
    "area": 0.407,  
    "bbox": [  
      0.02663142641129032,  
      0.40691584277841153,  
      0.9524163571731749,  
      0.42766634515266866  
    ]  
  },  
  {  
    "id": 2,  
    "category_id": 9,  
    "image_id": 2,  
    "area": 0.27,  
    "bbox": [  
      0.11803319477782331,  
      0.41586723392402375,  
      0.7765206955096307,  
      0.3483334397217212  
    ]  
  },  
  ...  
],
```

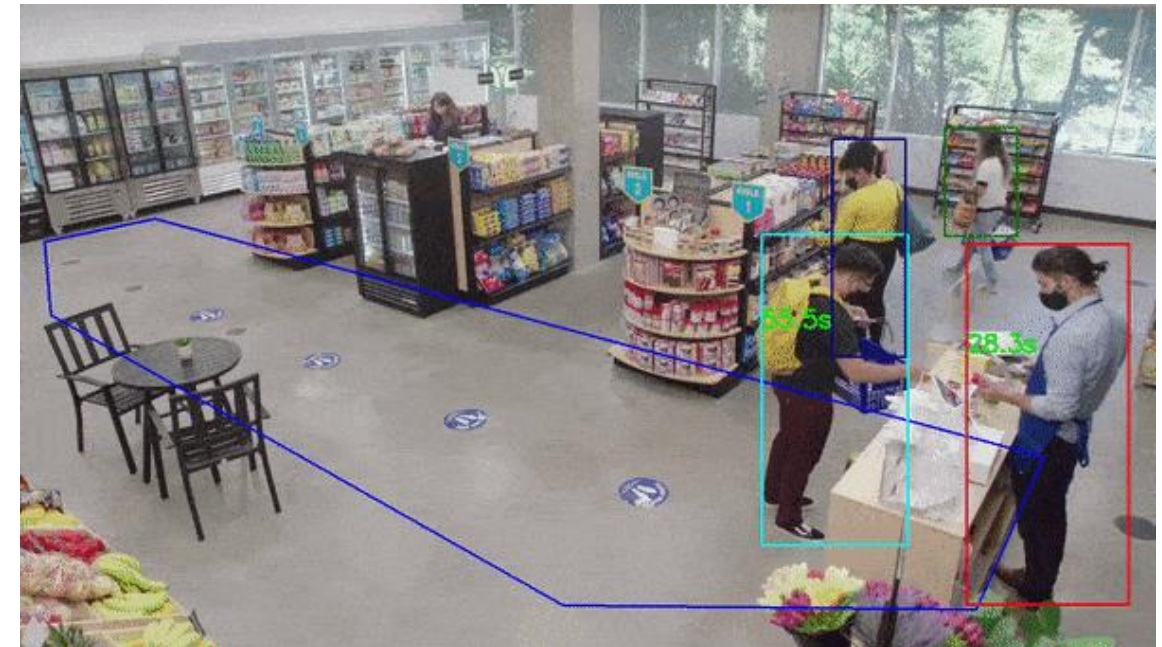
```
"categories": [  
  {  
    "id": 1,  
    "name": "vegall original mixed vegetables"  
  },  
  {  
    "id": 2,  
    "name": "Amy's organic soups lentil vegetable"  
  },  
  {  
    "id": 3,  
    "name": "Arrowhead 8oz"  
  },  
  ...  
]
```


Video Analysis

- Person counting



- Entrance Counting
 - wait times for a checkout line
 - measure foot traffic in a lobby or a specific floor



Video Analysis

- Social distancing and face mask detection



- Video Retrieval
 - Create a search index
 - Add documents (videos and images) to it
 - Search with natural language.

Azure AI Language

Extract information

Use Natural Language Understanding (NLU) to extract information from unstructured text. For example, identify key phrases or Personally Identifiable Information (PII), summarize text, recognize and categorize named entities, or customize an entity extraction model on top of your...

-  Extract key phrases
-  Find linked entities
-  Named Entity Recognition (NER)
-  Personally Identifiable Information (PII) detection
-  Custom Named Entity Recognition (custom NER)
-  Text analytics for health
-  Custom text analytics for health

Summarize text-based content

Summarize lengthy documents and conversation transcripts.

-  Document summarization
-  Conversation summarization

Classify Text

Use Natural Language Understanding (NLU) to detect the language or classify the sentiment of text you have. You can also classify your text documents by customizing a classification model over your dataset.

-  Analyze sentiment and mine text for opinions
-  Detect Language
-  Custom text classification

Answer questions

Provide answers to questions being asked in unstructured texts using our prebuilt capabilities, or customize your domain specific question-and-answer pairs over data you provide.

-  Question answering

Understand conversations

Create your own models to classify conversational utterances and extract detailed information from them to fulfill scenarios.

-  Conversational language understanding
-  Orchestration workflow

Translate text

Use cloud-based neural machine translation to build intelligent, multi-language solutions for your applications.

-  Use machine translation on text.

Azure AI Language Features

- Named Entity Recognition (NER)
 - Categorizes entities (words or phrases) in unstructured text across several predefined category groups. For example: people, events, places, dates, and more.
- Personally identifying (PII) and health (PHI) information detection
 - Identifies, categorizes, and redacts sensitive information in both unstructured text documents, and conversation transcripts. For example: phone numbers, email addresses, forms of identification, and more.
 - Customers can now redact transcripts, chats, and other text written in a conversational style (i.e. text with "um"s, "ah"s, multiple speakers etc.)

Event Entity value: trip Confidence: 74.00%	Location GPE Entity value: Seattle Confidence: 100.00%	DateTime DateRange Entity value: last week Confidence: 80.00%
--	---	--

Original text

I had a wonderful trip to Seattle last week.

trip to Seattle last week
Event Location DateTime

URL Entity value: www.contososteakhouse.com Confidence: 80.00%	PhoneNumber Entity value: 312-555-0176 Confidence: 80.00%	Email Entity value: order@contososteakhouse.com Confidence: 80.00%	Organization Entity value: contososteakhouse Confidence: 46.00%
--	--	--	---

Original text

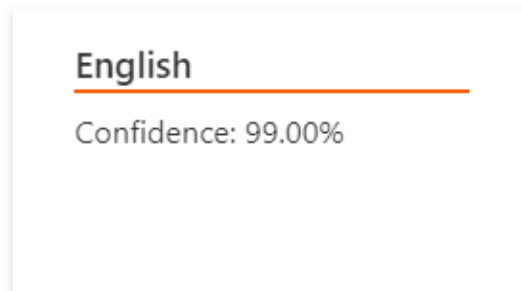
You can even pre-order from their online menu at www.contososteakhouse.com, call 312-555-0176 or send email to order@contososteakhouse.com!

www.contososteakhouse.com 312-555-0176 order@contososteakhouse.com
URL PhoneNumber Email

Azure AI Language Features

- Language and script detection
 - Detect the language a document is written in, and returns a language code for a wide range of languages, variants, dialects, and some regional/cultural languages.

Language detected

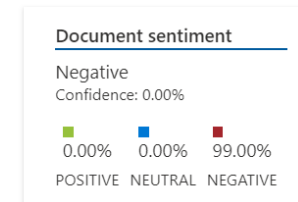


Original text

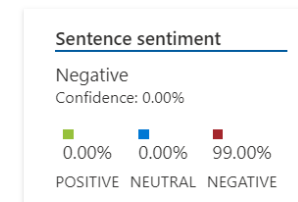
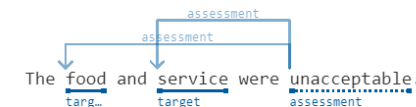
This document is in English.

- Sentiment Analysis and opinion mining

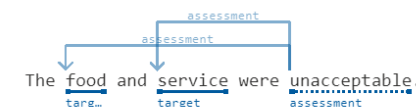
Analyzed sentiment



Sentence 1



Original text



Azure AI Language Features

- **Summarization:** Uses extractive text summarization to produce a summary of documents and conversation transcriptions. It extracts sentences that collectively represent the most important or relevant information within the original content.

Sentence 1

We're delighted to announce that Cognitive Service for Language service now supports extractive summarization!

Rank Score

100.00%

Sentence 2

Document summarization is a feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content.

Rank Score

39.00%

Sentence 3

Extractive summarization condenses articles, papers, or documents to key sentences.

Rank Score

96.00%

Original text

We're delighted to announce that Cognitive Service for Language service now supports extractive summarization! In general, there are two approaches for automatic document summarization: extractive and abstractive. This feature provides extractive summarization. Document summarization is a feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content. This feature is designed to shorten content that could be considered too long to read. Extractive summarization condenses articles, papers, or documents to key sentences.

- **Key phrase extraction**
 - Evaluates and returns the main concepts in unstructured text, and returns them as a list.

Key phrases

Pike place market, favorite Seattle attraction

Original text

Pike place market is my favorite Seattle attraction.

Key phrase

Key phrase

Azure AI Language Features

- Entity linking
 - Disambiguates the identity of entities (words or phrases) found in unstructured text and returns links to Wikipedia.

Linked entity Microsoft -Wikipedia	Linked entity Bill Gates -Wikipedia	Linked entity Paul Allen -Wikipedia	Linked entity April 4 -Wikipedia
---	--	--	---

Original text

Microsoft was founded by Bill Gates and Paul Allen on April 4, 1975.

- Text analytics for health
 - Extracts and labels relevant medical information from unstructured texts such as doctor's notes, discharge summaries, clinical documents, and electronic health records.

Ribavirin UMLS: C0035525 was also evaluated against SARS-CoV-2 infection , but the antiviral UMLS: C0003451

property of drugs UMLS: C0013227 is still not well established against the SARS-CoV-2 UMLS: C5203670 negation .

In addition, after oral administration, the drug was rapidly absorbed into the GI tract UMLS: C0017189 .

The drug has oral bioavailability around 64 % with large volume of distribution.

Text analytics for health

Named entity recognition

Ribavirin (UMLS: C0035525) was also evaluated against SARS-CoV-2 infection (UMLS: C0205462), but the antiviral (UMLS: C0003451) property of drugs (UMLS: C0013227) is still not well established against the SARS-CoV-2 (UMLS: C5203670) negation.

In addition, after oral administration, the drug was rapidly absorbed into the GI tract (UMLS: C0017189).

The drug has oral bioavailability around 64% with large volume of distribution.

Supported entity types.

Relation extraction

Woman (UMLS: C0043210) in NAD (UMLS: C2051415) with a h/o CAD (UMLS: C1956346), DM2 (UMLS: C0011860), asthma (UMLS: C0004096) and HTN (UMLS: C0020538) on ramipril (UMLS: C0072973) for 8 years awoke from sleep around 2:30 am this morning of a sore throat (UMLS: C0242429) and swelling of tongue (UMLS: C0236068). She came immediately to the ED b/c she was having difficulty swallowing (UMLS: C0011168) and some trouble breathing (UMLS: C0013404) due to

Supported relation types.

Entity linking (to UMLS – Unified Medical Language System)

patients must have histologically confirmed r / r NHL or HL (defined by who criteria). patients with chronic lymphocytic leukemia (cll) and small lymphocytic lymphoma (sll) are eligible. in addition patients with NHL other than diffuse large b cell lymphoma (dlbcl) must have received at least 2 prior therapies. patients with diagnosed with HL are eligible if there is no available

Assertion detection

Patient's brother died Someone Else at the age of 64 Someone Else from lung cancer Someone Else. She was admitted for likely gastroparesis Positive Possible but remains unsure if she wants to start adjuvant hormonal therapy Hypothetical. Please hold lactulose Negative Conditional if diarrhea Hypothetical worsen.

Supported assertions

Azure AI Language Features

- Question answering
 - Finds the most appropriate answer for inputs from your users, and is commonly used to build conversational client applications, such as social media applications, chat bots, and speech-enabled desktop applications.
 - QnA Maker

Where can I go to charge my laptop?

Short answer

you can use one of the power sockets available by one of the sofa seating areas located across the mall

Long answer

I'm sorry, we do not have laptop chargers available. If you have your charger, you can use one of the power sockets available by one of the sofa seating areas located across the mall.

Type your message and press enter



- Custom Named Entity Recognition (Custom NER)
 - Build custom AI models to extract custom entity categories (labels for words or phrases), using unstructured text that you provide.

Date Entity value: 6/29/2018 Confidence: 100.00%	BorrowerName Entity value: Parker McLean Confidence: 100.00%	BorrowerAddress Entity value: 1234 Main Rd Confidence: 96.00%	BorrowerCity Entity value: Frederick Confidence: 100.00%	BorrowerState Entity value: Nebraska Confidence: 100.00%
LenderName Entity value: Dylan Williams Confidence: 100.00%	LenderAddress Entity value: 7890 May Street Confidence: 100.00%	LenderCity Entity value: Winchester Confidence: 100.00%	LenderState Entity value: Kentucky Confidence: 100.00%	

Original text

Date 6/29/2018
Date

This is a Loan agreement between the two individuals mentioned below in the parties section of the agreement.

I. Parties of agreement

- Parker McLean with a mailing address of 1234 Main Rd, City of Frederick, State of Nebraska, (the Borrower)

- Dylan Williams with a mailing address of 7890 May Street, City of Winchester, State of Kentucky, (the "Lender")

<http://corenlp.run/>
<https://tagme.d4science.org/tagme/>

Input Text

Italiano

English

Deutsche

Trump is the 45th and current president of the United States. Modi, prime minister of India, met Trump in 2019 and Obama in 2014.

Many links

Few links

Reset

TAGME!

Tagged text

Topics

Trump is the 45th and current president of the United States. Modi, prime minister of India, met

Donald Trump

Donald John Trump (born June 14, 1946) is an American businessman, politician, television personality, author, and candidate for the Republican nomination for President of the United States in the 201...

Narendra Modi

Narendra Damodardas Modi (, born 17 September 1950) is the 15th and current Prime Minister of India, in office since 26 May 2014. Modi, a leader of the Bharatiya Janata Party was the Chief Minister of...

— Text to annotate —

The quick brown fox jumped over the lazy dog.

— Annotations —

parts-of-speech x

named entities x

dependency parse x

openie x

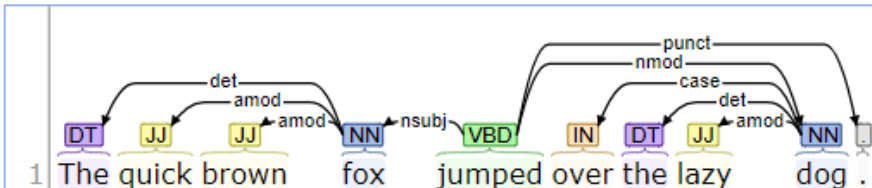
Part-of-Speech:

1 The quick brown fox jumped over the lazy dog .

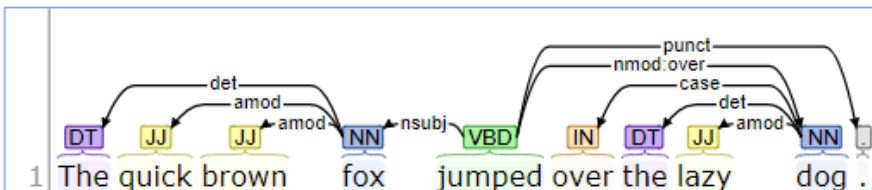
Named Entity Recognition:

1 The quick brown fox jumped over the lazy dog .

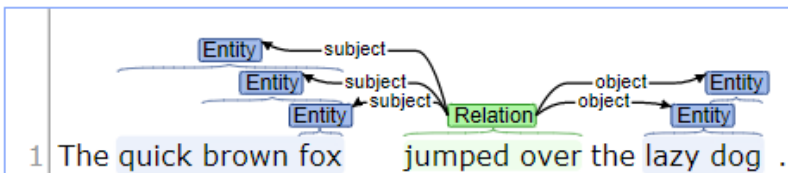
Basic Dependencies:



Enhanced++ Dependencies:



Open IE:



<http://corenlp.run/>

— Text to annotate —

Trump is president of U.S.A. Mr. Modi met him in 2019.

— Annotations —

☐ parts-of-speech ☒ named entities ☒ dependency parse ☒ openie ☒

Part-of-Speech:

1	Trump	is	president	of	U.S.A.	.	
2	Mr.	Modi	met	him	in	2019	.

Named Entity Recognition:

1	PERSON	TITLE	COUNTRY	
2	PERSON	DATE		

Basic Dependencies:

1	Trump	is	president	of	U.S.A.	.	
2	Mr.	Modi	met	him	in	2019	.

Open IE:

1	Entity	subject	Relation	object	Entity	Entity
2	Entity	subject	Relation	object	Entity	Entity

Content Moderation

Image moderation

Enhance your ability to detect potentially offensive or unwanted images through machine learning-based classifiers, custom lists and optical character recognition (OCR).

Text moderation

Use content filtering to detect potential profanity in more than 100 languages, flag text that may be deemed inappropriate depending on context (in public preview) and match text against your custom lists. Content Moderator also helps check for personally identifiable information (PII).

Video moderation

Enable machine-assisted detection of possible adult and racy content in videos. The video moderation service (in public preview) is available as part of Azure Media Services.

Human review tool

The best content moderation results come from humans and machines working together. Use the review tool when prediction confidence can be improved or tempered with a real-world context.

Azure AI Document Intelligence

- Quickly extract text, key-value pairs, tables and structure from documents



Simple text extraction

Easily pull data and organize information with prebuilt and custom features—no manual labeling required.



Customized results

Get output tailored to your layouts with automatic custom extraction and improve it with human feedback.



Flexible deployment

Ingest data from the cloud or at the edge and apply to search indexes, business automation workflows, and more.



Built-in security

Rely on enterprise-grade security and privacy applied to both your data and any trained models.



Analyze forms and documents

Make data-driven decisions by extracting data from documents and putting it into your data visualization service for analysis.



Create intelligent search indexes

Easily find specific information in your documents and forms, such as total accounts payable, by integrating AI Document Intelligence with Azure Applied AI Search.

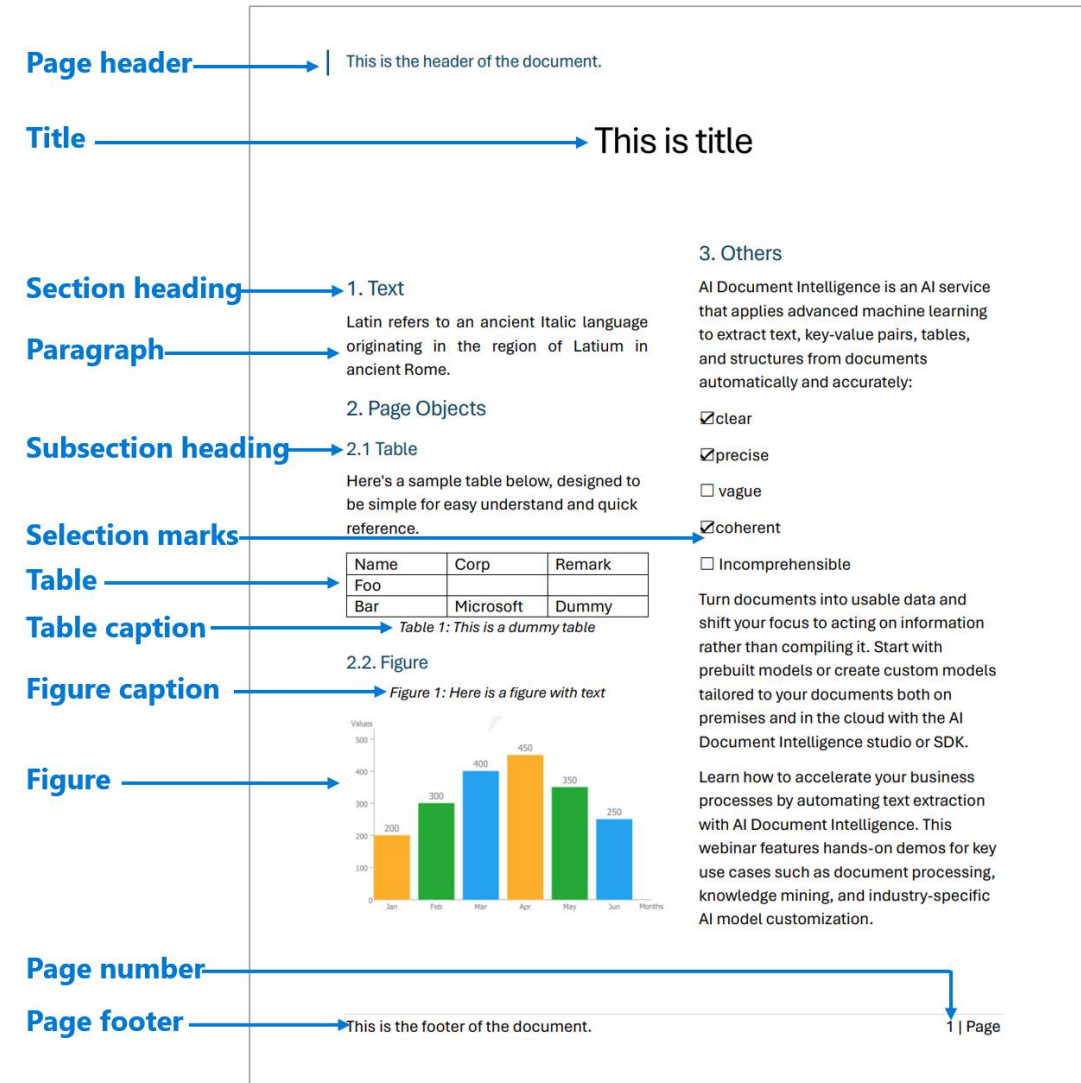


Automate business workflows

Extract text, key-value pairs, tables, and structures from documents, and pipe them into your back-end systems to perform tasks such as claim, invoice, and receipt processing.

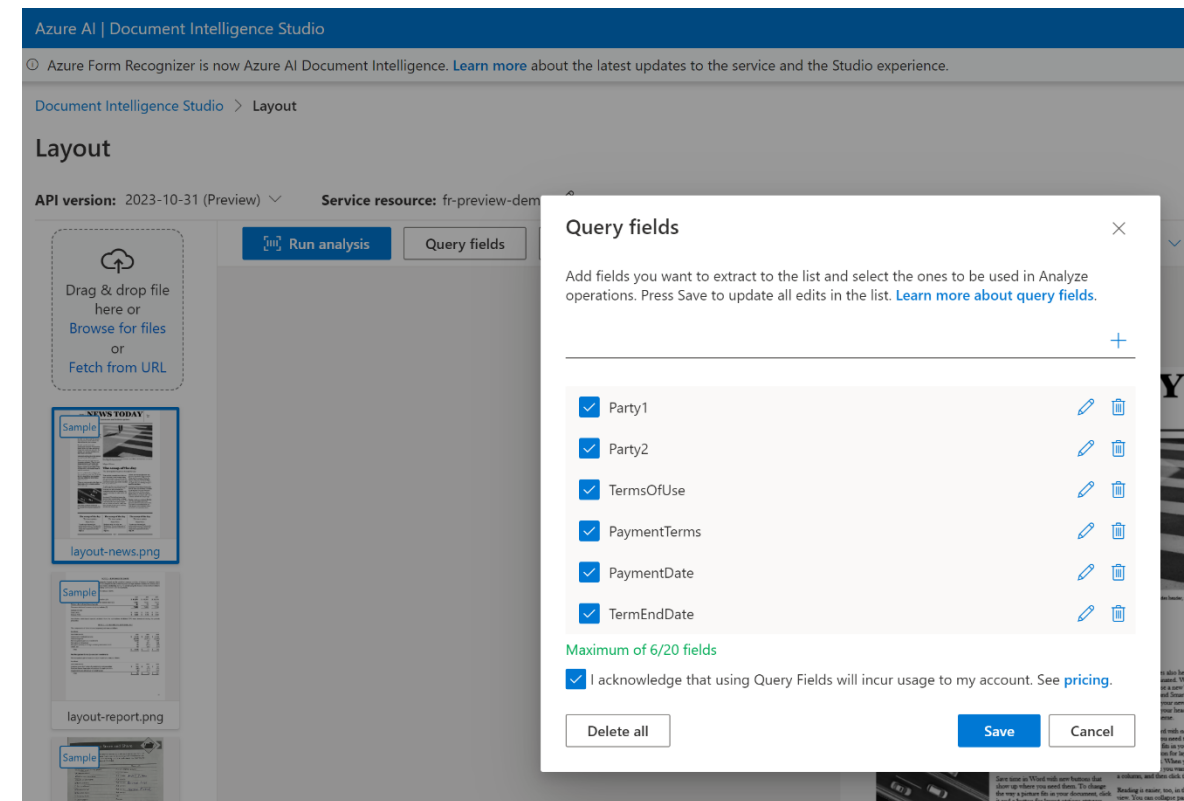
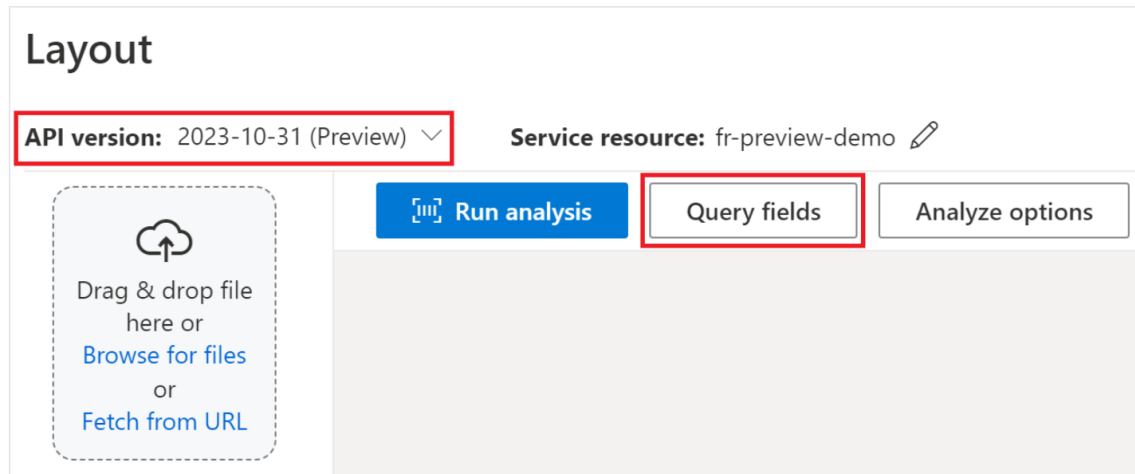
Document Intelligence read and layout model

- Optical Character Recognition (OCR) for documents is optimized for large text-heavy documents in multiple file formats and global languages.
- Paragraph extraction
 - Text block, bounding polygon coordinates.
- Text, lines, and words extraction
 - Extracts print and handwritten style text as lines and words.
 - Bounding polygon coordinates and confidence for the extracted words.
 - Spans pointing to the associated text.
- Formulas, such as mathematical equations
- All font properties of text
 - similarFontFamily, fontStyle for styles such as italic and normal, fontWeight for bold or normal, color for color of the text, and backgroundColor for color of the text bounding box.
- Barcode property extraction
- Language detection
- Convert an analog PDF, such as scanned-image PDF files, to a PDF with embedded text.



Query Field Extraction

- specify the fields you want to extract and Document Intelligence analyzes the document accordingly.
- You can pass a list of field labels like Party1, Party2, TermsOfUse, PaymentTerms, PaymentDate, and TermEndDate as part of the analyze document request.



CONTOSO LTD.

Contoso Headquarters
123 456th St.
New York, NY, 10001

Microsoft Corp
123 Other St,
Redmond WA, 98052

INVOICE

SALESPERSON	REQUESTOR	SHIPPED VIA	F.O.B. POINT	TERMS
Jessica Finance 123 Bill St, Redmond WA, 98052	Microsoft Licensey 123 Ship St, Redmond WA, 98052			

DATE	ITEM CODE	DESCRIPTION	QTY	UM	PRICE	TAX	AMOUNT
3/4/2021	A123	Consulting Services	2	hours	\$30.00	\$6.00	\$40.00
3/5/2021	B456	Document Fee	3		\$10.00	\$3.00	\$30.00
3/6/2021	C789	Printing Fee	10	pages	\$1.00	\$1.00	\$10.00

THANK YOU FOR YOUR BUSINESS!

REMIT TO:
Carlinson Billing
123 Remit St
New York, NY, 10001

Fields Content Result Code

Prebuilt invoice

Key-Value pairs

- INVOICE: 92.40%
- INV-100
- INVOICE DATE: 90.80%
- 11/15/2019
- DUE DATE: 90.70%
- 12/15/2019
- CUSTOMER NAME: 88.90%
- MICROSOFT CORPORATION
- SERVICE PERIOD: 87.40%
- 10/14/2019 – 11/14/2019
- CUSTOMER ID: 90.70%
- CID-12345

Prebuilt Doc Intelligence Extraction Models

Receipt

Fields **Result** **Code**

DocType: receipt.hotel

- ArrivalDate #1 99.40%
2021-03-27
- Currency 99.50%
USD
- DepartureDate #1 99.30%
2021-03-28
- Items (6) #1
- MerchantAddress #1 98.70%
5600 148th Ave NE, Redmond, WA 98052
HouseNumber
5600

Check

Fields **Result** **Code**

PayerAddress #1 99.50%
123 Main St, Redmond, WA 98052
HouseNumber
123
Road
Main St
PostalCode
98052
City

NumberAmount
Content \$ 123,456.00
Value 123456
Confidence 99.50%

PayerName #1 99.40%
Contoso Ltd.

PayTo #1 99.50%
22nd Century Insurance John Doe

WordAmount #1 99.50%
123456

Identity (ID)

Fields **Result** **Code**

DocType: idDocument.passport

- CountryRegion #1 99.00%
MYS
- DateOfBirth #1 99.00%
1975-11-02
- DateOfExpiration #1 99.00%
2022-03-10
- DateOfIssue #1 99.00%
2017-03-11
- DocumentNumber #1 99.00%
R00000000
- DocumentType #1 99.00%

Prebuilt Doc Intelligence Extraction Models

Pay stub

CONTOSO LTD

EARNINGS	HOURS	RATE	CURRENT	YTD
Regular Hours	56.00	24.00	\$ 1,344.00	\$ 5,000.00
Overtime	2.00	36.00	\$ 72.00	\$ 500.00
Sick Pay	8.00	24.00	\$ 192.00	\$ 500.00
Vacation Pay	8.00	24.00	\$ 192.00	\$ 500.00
Holiday Pay	8.00	24.00	\$ 192.00	\$ 500.00
	82.00		\$ 1,992.00	\$ 7,000.00

DEDUCTIONS	RATE	CURRENT	YTD
Aftertax Health Ins		\$ 120.00	\$ 500.00
Aftertax Dental Ins		\$ 35.00	\$ 400.00
Aftertax 401k		\$ 78.00	\$ 200.00
Child Support		\$ 155.00	\$ 200.00
Garnishment		\$ 45.00	\$ 500.00
		\$ 433.00	\$ 1,800.00

TAXES	RATE	CURRENT	YTD
Federal Income Tax		\$ 100.00	\$ 500.00
Social Security	0.0620	\$ 123.50	\$ 475.22
Medicare	0.0145	\$ 28.88	\$ 199.00
State Income Tax		\$ 50.00	\$ 200.00
Local Income Tax		\$ 10.00	\$ 40.00
		\$ 312.39	\$ 1,414.22

NET PAY	CURRENT	YTD
	\$ 1,246.61	\$ 3,785.78

Direct Deposited to Account of:
Carl Anderson

Routing Number:
485066128

Account Number:
8300567112

Amount:
\$ 1,246.61

CONTOSO LTD

123 Main Street
Redmond, WA 98053

1/12/2024

1001

EARNINGS STATEMENT

Pay Date
1/12/2024

Pay Start
1/1/2024

Pay End
1/12/2024

Carl Anderson

203 Denver Blvd
Seattle, WA 98046

SSN: 997-65-1234

DOB: 01-09-1997

CurrentPeriodGrossPay #1

99.509

CurrentPeriodNetPay #1

99.509

1246.61

CurrentPeriodTaxes #1

99.509

312.39

EmployeeAddress #1

99.509

203 Denver Blvd Seattle, WA 98040

HouseNumber

203

Road

Denver Blvd

PostalCode

98040

City

Seattle

State

WA

StreetAddress

203 Denver Blvd

Bank statement

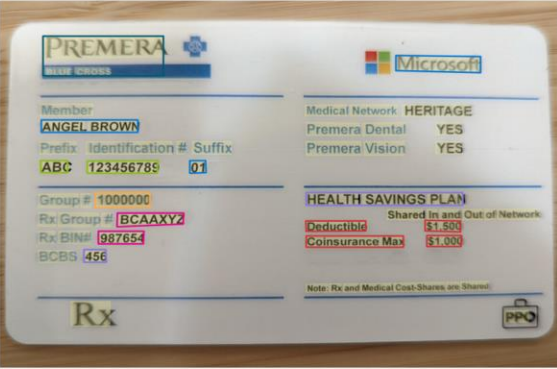
CONTOSO BANK		Account Summary	
Account Number 168552031585	Account Type Simple Checking	Statement Period Nov 1, 2023 - Nov 30, 2023	
Lela R. Duvall 2096 Godfrey Road New York, NY 10021			
Beginning Balance on Nov 1, 2023	+ 2,500.00		
Deposits / Credits	- 1,200.00		
Service Fees	- 10.00		
Ending Balance on Nov 30, 2023	\$ 4,290.00		
Deposits / Credits			
Date	Description	Amount	
11/01/2023	Deposit	\$ 2,500.00	
11/02/2023	Deposit	\$ 2,500.00	
Total Deposits / Credits		\$ 2,500.00	
Withdrawals / Debits			
Date	Description	Amount	
11/01/2023	Online Payment	\$ 1,000.00	
11/02/2023	Online Transfer to xxxxxxxx1396	\$ 1,200.00	
Total Withdrawals / Debits		\$ 1,200.00	
Service Fees			
Date	Description	Amount	
11/01/2023	International Transaction Fee	\$ 10.00	
Total Service Fees		\$ 10.00	
Classified as Microsoft Confidential		Page 1 of 1	

Prebuilt Doc Intelligence Extraction Models

Health insurance card

Run analysis

Analyze options



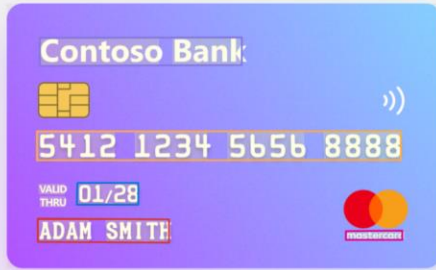
Fields	Result	Code
Copays (2)	#1	
1 Amount	\$1,500	
Benefit		
Deductible		
2 Amount	\$1,000	
Benefit		
Coinsurance Max		
GroupNumber	#1	99.50%
1000000		
IdNumber	#1	99.50%
Number	123456789	99.50%
Prefix	ABC	99.50%
Insurer	#1	99.50%
PREMERA BLUE CROSS		

Credit card model

Run analysis

Query fields

Analyze options



Fields	Result	Code
DocType: creditCard		
CardHolderName	#1	99.50%
ADAM SMITH		
CardNumber	#1	99.50%
5412 1234 5656 8888		
CardVerificationValue	#1	99.50%
123		
CustomerServicePhoneNumbers (2)	#1	
1 +1 200-345-6789		
2 +1 200-000-8888		
ExpirationDate	#1	99.50%
01/28		
IssuingBank	#1	99.50%
Contoso Bank		
PaymentNetwork	#1	99.10%
mastercard		

Contract model

Analyze

All pages

Range



Fields	Content	Result	Code
DocType: contract			
EffectiveDate	#1	99.99%	
15 day of October, 2022			
ExecutionDate	#1	99.99%	
15 day of October, 2022			
Jurisdictions	#1	100.00%	
Clause			
This Agreement shall be governed by and construed in accordance with the internal laws of the State of Washington applicable to agreements made and to be performed entirely within such state.			
Region		100.00%	
Washington			
Parties (2)	#1		
Contoso Corporation			
Adaptive Works Cycles			
Title	#1	99.98%	
WEB HOSTING AGREEMENT			

Prebuilt Doc Intelligence Extraction Models

Marriage certificate model

Run analysis

Query fields

Analyze options

Fields Result Code

DocType: marriageCertificate.us

IssueDate #1	99.50%
March 22, 2017	
MarriageDate #1	98.60%
19th March AD, 2017	
MarriagePlace #1	99.50%
Detroit Wayne MICHIGAN,	

US mortgage 1003 form

Run analysis

Query fields

Analyze options

Fields Result Code

DocType: mortgage.us.1003

AgencyCaseNumber #1	99.50%
115894	
Borrower #1	
BirthDate	99.50%
04 / 07 / 1989	
CellPhoneNumber	90.20%
(831) 728 - 4766	
NumberOfBorrowers	99.50%
2	
CurrentAddress	99.50%
1634 W Glenoaks Blvd Glendale CA 91201 United States	
HouseNumber	
1634	
Road	
W Glenoaks Blvd	

US Tax W-2 model

Analyze

All pages

Range

Fields Result Code

AllocatedTips #1	99.90%
874.2	
ControlNumber #1	99.90%
000086242	
DependentCareBenefits #1	99.90%
9873.2	
Employee #1	
Address	99.90%
4567 MAIN STREET BUFFALO, WA 12345	
HouseNumber	
4567	
Road	
MAIN STREET	
PostalCode	
12345	

Azure AI Speech



Transcribe speech to text

Transcribe call center or meeting conversations. Go global with audio-captioning in more than 100 languages.

[Learn more](#)



Transcribe audio with OpenAI Whisper

Transform your call centers using the latest OpenAI Whisper model in Azure AI Speech or Azure OpenAI Service.

[Read the blog](#)



Verify and recognize speakers

Confirm a person's identity or recognize who's speaking in a meeting by adding speaker verification and identification to your app.

[Learn more](#)



Convert text to speech

Build bots that speak naturally. Differentiate your brand with customized, realistic voices and speaking styles.

[Learn more](#)



Build custom voices

Build natural-sounding voices with custom neural voice.

[Learn more](#)



Enable multilingual communication

Translate audio or video data from and into an ever-growing list of supported languages. Customize translations to your industry.

[Learn more](#)



Speech analytics

Analyze audio or video call recordings to gain deep insights. Summarize key topics and extract or redact personal identification information.

[Learn more](#)



Build your avatars

Bring your brand to life using pre-built or custom avatars with natural-sounding voices.

[Learn more](#)



Embedd speech

Use embedded speech to power on-device speech to text and text to speech scenarios where cloud connectivity is intermittent or unavailable.

[Learn more](#)

Speech to text

- Applications
 - Transcriptions, captions, or subtitles for live meetings: Real-time audio transcription for accessibility and record-keeping.
 - Diarization: Identifying and distinguishing between different speakers in the audio.
 - Pronunciation assessment: Evaluating and providing feedback on pronunciation accuracy.
 - Call center agents assist: Providing real-time transcription to assist customer service representatives.
 - Dictation: Transcribing spoken words into written text for documentation purposes.
 - Voice agents: Enabling interactive voice response systems to transcribe user queries and commands.
- Caption and speech synchronization: synchronize captions with the audio track, whether it's in real-time or with a prerecording.
- Get partial results: Speech recognition results are subject to change while an utterance is still being recognized.
- Language identification: identify languages spoken in audio (once or continuous)
- Multilingual speech to speech.
- The Speech service supports output formats such as SRT (SubRip Text) and WebVTT (Web Video Text Tracks)

```
00:00:00,180 --> 00:00:03,230
Welcome to applied Mathematics course 201.
```

Text to Speech

- Real-time speech synthesis
- Asynchronous synthesis of long audio
- Speech Synthesis Markup Language (SSML)
 - Used to customize text to speech outputs.
 - Adjust pitch, add pauses, improve pronunciation, change speaking rate, adjust volume, and attribute multiple voices to a single document.
- Visemes
 - Key poses in observed speech, including the position of the lips, jaw, and tongue in producing a particular phoneme.
 - Visemes can be used to generate facial animation data
- Text to speech avatar (speech as well as video)

	Gender	Female	Male	Neutral										
	Age group	Child	Young adult	Adult	Senior									
	Capability	Context-aware	Multi-lingual	Multi-style										
	Recommendation	Featured	Popular											
	Speaking style	Advertisement - upbeat	Affectionate	Angry	Assistant	Calm	Chat	Chat - casual	Cheerful	Conversation	Customer serv	More		
	Personality	Animated	Approachable	Authentic	Authoritative	Bright	Calm	Caring	Casual	Cheerful	Clear	Confident	Cris	More
	Tailored scenario	Advertisement	Assistant	Audiobook	Audiobooks	Chat	Documentary	E-learning	Gaming	Meditation	Narration	More		

Pronunciation assessment

Parameter	Description	Granularity
AccuracyScore	How closely the phonemes match a native speaker's pronunciation. Syllable, word, and full text accuracy scores are aggregated from the phoneme-level accuracy score, and refined with assessment objectives.	Phoneme level, Syllable level (en-US only), Word level, Full Text level
FluencyScore	How closely the speech matches a native speaker's use of silent breaks between words.	Full Text level
CompletenessScore	Completeness of the speech, calculated by the ratio of pronounced words to the input reference text.	Full Text level
ProsodyScore	How natural the given speech is, including stress, intonation, speaking speed, and rhythm.	Full Text level
PronScore	Overall score of the pronunciation quality of the given speech, calculated from AccuracyScore, FluencyScore, CompletenessScore, and ProsodyScore with weight	Full Text level
ErrorType	- Whether a word is omitted, inserted, or improperly inserted with a break. - Indicates a missing break at punctuation. - Indicates whether a word is badly pronounced, or monotonically rising, falling, or flat on the utterance. Mispronunciation when the pronunciation AccuracyScore for a word is below 60.	Word level
GrammarScore	Correctness in using grammar and variety of sentence patterns. Lexical accuracy, grammatical accuracy, and diversity of sentence structures jointly elevate grammatical errors.	Full Text level
TopicScore	Level of understanding and engagement with the topic	Full Text level

Azure OpenAI service

- Provides REST API access to OpenAI's powerful language models including o1-preview, o1-mini, GPT-4o, GPT-4o mini, GPT-4 Turbo with Vision, GPT-4, GPT-3.5-Turbo, and Embeddings model series.
- Models available
 - o1-preview & o1-mini
 - GPT-4o & GPT-4o mini
 - GPT-4o audio
 - GPT-4 series (including GPT-4 Turbo with Vision)
 - GPT-3.5-Turbo series
 - Embeddings series (text-embedding-3-large, text-embedding-3-small, text-embedding-ada-002)
 - DALL·E, Whisper.
- Fine-tuning
 - GPT-4o-mini (preview)
 - GPT-4 (preview)
 - GPT-3.5-Turbo (0613)
 - babbage-002
 - davinci-002.

Model ID	Max request (tokens)	Training Data (up to)
babbage-002	16,384	Sep 2021
davinci-002	16,384	Sep 2021
gpt-35-turbo (0613)	4,096	Sep 2021
gpt-35-turbo (1106)	Input: 16,385 Output: 4,096	Sep 2021
gpt-35-turbo (0125)	16,385	Sep 2021
gpt-4 (0613) ¹	8192	Sep 2021
gpt-4o-mini ¹ (2024-07-18)	Input: 128,000 Output: 16,384 Training example context length: 64,536	Oct 2023
gpt-4o ¹ (2024-08-06)	Input: 128,000 Output: 16,384 Training example context length: 64,536	Oct 2023

Safety system for monitoring content generated by both LLMs and humans.



Moderate text content

Run moderation tests on text contents. Assess the test results with detected severities. Experiment with different threshold levels.



Groundedness detection

Groundedness detection detects ungroundedness generated by the large language models (LLMs).



Protected material detection (Text)

Use protected material detection to detect and protect third-party text material in LLM output.



Prompt Shields

Prompt Shields provides a unified API that addresses Jailbreak attacks and Indirect attacks.



Moderate image content

Run moderation tests on image contents. Assess the test results with detected severities. Experiment with different threshold levels.



Moderate multimodal content

Run moderation tests on image and text combined contents. Assess the test results with detected severities.



Custom categories

Create and manage content categories specific to your needs for enhanced moderation and filtering.



Safety system message

Use the framework of metaprompt that helps you potentially mitigate different types of harm.

Search APIs

Bing Autosuggest

Help users complete queries faster by adding intelligent type-ahead capabilities

Bing Custom Search

Create a custom search engine with ad-free results.

Bing Entity Search

Recognize and classify named entities and find search results based on them.

Bing Image Search

Add a variety of search image options to your app.

Bing News Search

Turn any app into a news search resource.

Bing Spell Check

Help users identify and fix word breaks, slang, names, homonyms and brands.

Bing Video Search

Add advanced video search features to your app.

Bing Visual Search

Enable users to search using images.

Bing Web Search

Enable safe, ad-free, location-aware search for your users, surfacing relevant information from [web results](#), [images](#), [news](#), [videos](#) and [visuals](#).

AutoSuggest

<https://www.microsoft.com/en-us/bing/apis/bing-autosuggest-api>

See it in action



Market

Preview

JSON

narendra modi
narendra modi news
narendra
narendra jha
narendra modi twitter
narendra modi movie
narendra modi biopic
narendra modi information

Thanks!!

Questions?